



ASSIGNMENT-1

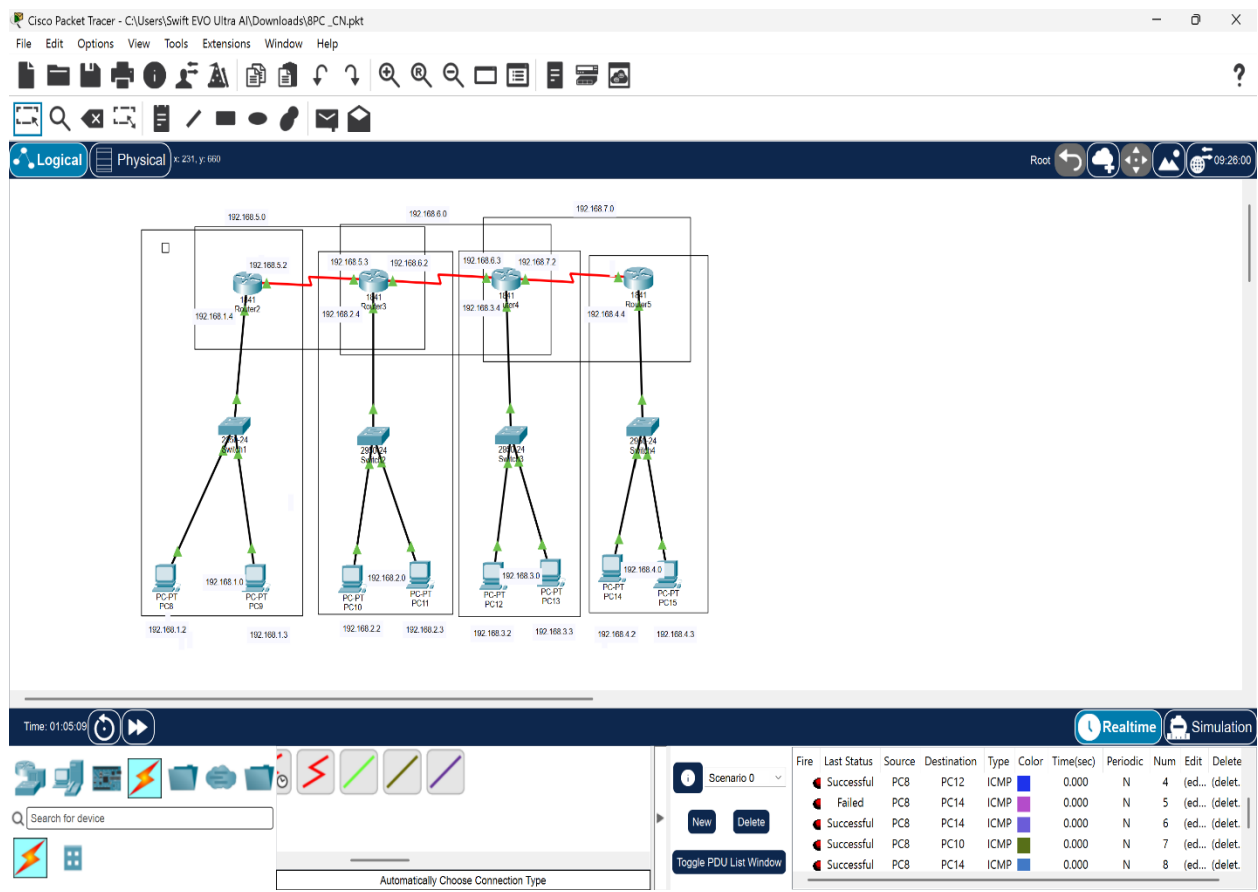
SUBJECT - COMPUTER NETWORK

CODE – 24CST-339

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Q1. Develop a Static Routing topology using 4 Routers, 4 Switches, and 8 PCs; assign IP addresses to all devices, configure static routes between networks, and clearly label all LAN/WAN Network IDs and IP addresses in text format.



Network IDs

- 1. LAN 1 Network ID: 192.168.1.0/24**
- 2. LAN 2 Network ID: 192.168.2.0/24**
- 3. LAN 3 Network ID: 192.168.3.0/24**
- 4. LAN 4 Network ID: 192.168.4.0/24**
- 5. Router2 ↔ Router3 Network ID: 192.168.5.0/24**
- 6. Router3 ↔ Router4 Network ID: 192.168.6.0/24**
- 7. Router4 ↔ Router5 Network ID: 192.168.7.0/24**

PC IP Addresses

Device IP Address

<u>PC8</u>	<u>192.168.1.2</u>
<u>PC9</u>	<u>192.168.1.3</u>
<u>PC10</u>	<u>192.168.2.2</u>
<u>PC11</u>	<u>192.168.2.3</u>
<u>PC12</u>	<u>192.168.3.2</u>
<u>PC13</u>	<u>192.168.3.3</u>
<u>PC14</u>	<u>192.168.4.2</u>
<u>PC15</u>	<u>192.168.4.3</u>

Router Interface IP Addresses

Router2

- LAN Interface: 192.168.1.4
- Serial Interface to Router3: 192.168.5.2

Router3

- LAN Interface: 192.168.2.4
- Serial Interface to Router2: 192.168.5.3
- Serial Interface to Router4: 192.168.6.2

Router4

- LAN Interface: 192.168.3.4
- Serial Interface to Router3: 192.168.6.3
- Serial Interface to Router5: 192.168.7.2

Router5

- LAN Interface: 192.168.4.4
- Serial Interface to Router4: 192.168.7.3 (should be used; the diagram does not show it clearly)

Default Gateways

- LAN 1 Gateway: 192.168.1.4
- LAN 2 Gateway: 192.168.2.4

- LAN 3 Gateway: 192.168.3.4
- LAN 4 Gateway: 192.168.4.4

These are the network IDs and IP addresses in text format for the static routing topology.

Q2. Write the Difference Between Hub, Switch & Router in Tabular Format. Explain Why Router is the Most Intelligent and Hub is the Least Intelligent Device. (3 Marks)

Difference Between Hub, Switch and Router

ANSWER-

<u>Feature</u>	<u>Hub</u>	<u>Switch</u>	<u>Router</u>
<u>Definition</u>	<u>A hub is a basic networking device that connects multiple devices and broadcasts data to all ports.</u>	<u>A switch connects devices within a LAN and forwards data only to the intended device.</u>	<u>A router connects different networks and routes data packets between them.</u>
<u>OSI Layer</u>	<u>Physical Layer (Layer 1)</u>	<u>Data Link Layer (Layer 2)</u>	<u>Network Layer (Layer 3)</u>
<u>Data Forwarding</u>	<u>Broadcasts data to all connected devices</u>	<u>Sends data only to the destination device</u>	<u>Routes data to the correct network</u>

<u>Feature</u>	<u>Hub</u>	<u>Switch</u>	<u>Router</u>
<u>Address Used</u>	<u>No address table</u>	<u>Uses MAC Address</u>	<u>Uses IP Address</u>
<u>Collision Domain</u>	<u>One collision domain for all ports</u>	<u>Separate collision domain for each port</u>	<u>Separate collision domain for each interface</u>
<u>Broadcast Domain</u>	<u>One</u>	<u>One (unless VLANs are used)</u>	<u>Different broadcast domains</u>
<u>Speed</u>	<u>Slow</u>	<u>Faster than Hub</u>	<u>Fastest and most efficient</u>
<u>Security</u>	<u>Very Low</u>	<u>Moderate</u>	<u>High</u>
<u>Intelligence Level</u>	<u>Least Intelligent</u>	<u>Moderately Intelligent</u>	<u>Most Intelligent</u>
<u>Example Use</u>	<u>Small or old networks</u>	<u>LAN networks in offices and colleges</u>	<u>Internet and inter-network communication</u>

Why Router is the Most Intelligent Device

A router is considered the most intelligent networking device because:

1. It uses IP addresses to identify source and destination networks.
2. It makes routing decisions and selects the best path for data transmission.
3. It connects different networks, such as LANs, WANs, and the Internet.

4. It maintains routing tables to determine the optimal route.
5. It reduces unnecessary traffic by forwarding packets only to the required network.
6. It provides advanced features such as NAT, DHCP, Access Control Lists (ACLs), firewall functions, and dynamic routing protocols.

Example

When a computer in 192.168.1.0/24 wants to communicate with a computer in 192.168.4.0/24, the router analyzes the destination IP address and forwards the packet through the correct route.

Why Hub is the Least Intelligent Device

A hub is considered the least intelligent networking device because:

1. It does not maintain any address table.
2. It cannot identify the destination device.
3. It broadcasts every received packet to all ports.
4. It creates unnecessary network traffic.
5. It causes more collisions and reduces network efficiency.
6. It has no security or filtering capability.

Example

If PC1 sends data to PC2, the hub sends the same data to all connected devices, even though only PC2 needs it. All other devices receive and then discard the packet.

Conclusion

A Hub simply broadcasts data without making any decisions, making it the least intelligent device. A Switch improves network efficiency by using MAC addresses to forward data only to the intended device. A Router is the most intelligent device because it uses IP addresses, routing tables, and advanced networking features to connect and manage communication between different networks efficiently and securely.

Q3. What are Random Access Protocols? Why do we need them? Explain each category in detail with diagram and example. Also explain the difference between CSMA/CD and CSMA/CA. (3 Marks)

ANSWER-

What are Random Access Protocols?

Random Access Protocols are Medium Access Control (MAC) protocols that allow multiple devices to share the same communication channel. In these protocols, any device can transmit data whenever it has data to send without waiting for a fixed schedule.

Since multiple devices may attempt to transmit at the same time, collisions can occur. Random access protocols provide mechanisms to detect, avoid, or recover from collisions.

Need for Random Access Protocols

1. Allow multiple devices to share a common communication medium.
2. Improve utilization of network resources.
3. Reduce communication delays.
4. Provide fair access to all devices.
5. Handle collisions efficiently in shared networks.

Categories of Random Access Protocols

1. ALOHA

ALOHA is the simplest random access protocol. A station transmits data whenever it is ready without checking whether the channel is free.

Diagram

Station A -----> Data

Station B -----> Data

Collision occurs if both transmit together

Example

If Computer A and Computer B send data at the same time, their packets collide. Both computers wait for a random time and retransmit.

Advantages

- Simple implementation
- No channel sensing required

Disadvantages

- High collision rate
 - Low efficiency
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2. Slotted ALOHA

Slotted ALOHA improves ALOHA by dividing time into slots. A device can transmit only at the beginning of a time slot.

Diagram

| Slot 1 | Slot 2 | Slot 3 | Slot 4 |

Data Transmission Starts Here

Example

If a station has data to send, it waits until the next slot begins and then transmits.

Advantages

- Fewer collisions than pure ALOHA
- Better efficiency

Disadvantages

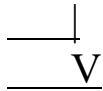
- Requires synchronization of all stations
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3. CSMA (Carrier Sense Multiple Access)

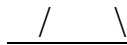
Before transmitting, a device listens to the channel. If the channel is busy, it waits. If the channel is free, it transmits.

Diagram

Listen Channel



Is Channel Free?



Yes No



Transmit Wait

Example

A computer checks whether another device is already sending data. If the channel is free, it starts transmission.

Advantages

- Reduces collisions
- Better channel utilization

Disadvantages

- Collisions can still occur due to propagation delay
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4. CSMA/CD (Carrier Sense Multiple Access with Collision)

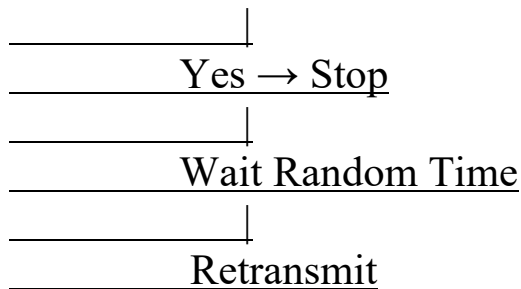
Detection)

Used mainly in traditional Ethernet networks.

A device listens before transmitting and continues monitoring the channel while transmitting. If a collision occurs, transmission stops immediately.

Diagram

Listen → Transmit → Detect Collision?

**Example**

Two computers begin transmitting simultaneously. Both detect the collision, stop transmission, wait for a random period, and then retransmit.

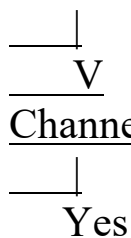
6. CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

Used mainly in wireless networks (Wi-Fi).

Since wireless devices cannot easily detect collisions, they try to avoid them by waiting and sending control messages before transmitting.

Diagram

Listen Channel



Send RTS

Receive CTS



Transmit Data

RTS = Request To Send

CTS = Clear To Send

Example

Before sending data, a Wi-Fi device requests permission from the access point. After receiving approval, it transmits.

Difference Between CSMA/CD and CSMA/CA

<u>Feature</u>	<u>CSMA/CD</u>	<u>CSMA/CA</u>
<u>Full Form</u>	<u>Carrier Sense Multiple Access with Collision Detection</u>	<u>Carrier Sense Multiple Access with Collision Avoidance</u>
<u>Working Principle</u>	<u>Detects collisions after they occur</u>	<u>Tries to prevent collisions before they occur</u>
<u>Used In</u>	<u>Wired Ethernet Networks</u>	<u>Wireless Wi-Fi Networks</u>
<u>Collision Handling</u>	<u>Detect and retransmit</u>	<u>Avoid and then transmit</u>
<u>Mechanism</u>	<u>Collision Detection</u>	<u>RTS/CTS and Backoff Mechanism</u>
<u>Efficiency</u>	<u>High in wired networks</u>	<u>Better for wireless environments</u>
<u>Transmission</u>	<u>Device transmits and monitors channel</u>	<u>Device waits before transmitting</u>
<u>Complexity</u>	<u>Simple</u>	<u>More complex</u>